

tave's own language, or using dynamically loaded modules written in C++, C, Fortran or other languages.

Octave has been built with MATLAB compatibility in mind, and shares many features with MATLAB. Some of these features include built-in support for complex numbers, powerful built-in math functions and extensive function libraries, and support for user-defined functions.

## TinyCAD

TinyCAD program lets users draw circuit diagrams in Windows. It has a full suite of features for designing circuit diagrams and integrating into a PCB layout program. It checks design rules, and generates parts lists and net lists for use with PCB layout programs. It also has an integral symbolic library editor, which makes adding and editing libraries effortless. Integration with Spice is possible using the Spice template engine.

TinyCAD allows users to design basic or complex circuit diagrams. With 755 symbols distributed in 42 libraries, it provides users a vast number of components that can be used directly in their designs. Each library can be edited by the users. It even allows users to add their own symbols and more libraries.

TinyCAD comes in a GUI platform that is very similar to many other circuit design software. The software comes with many useful tools that make it easier to design circuits. It automatically checks for duplicated references, unconnected items, unassigned reference designators and more. PCB export feature allows users to export the netlist of their circuit in order to create a printed circuit board.

Output formats supported by TinyCAD include TinyCAD, PADS-PCB, Eagle SCR, Protel, gEDA PCB and XML. Users can also export the image to clipboard and print the circuit.

## CEDAR Logic Simulator

CEDAR LS is an interactive digital logic simulator to teach logic design or test simple digital designs. It features both low-level logic gates as well as high-level components, including registers and a Z80 microprocessor emulator.

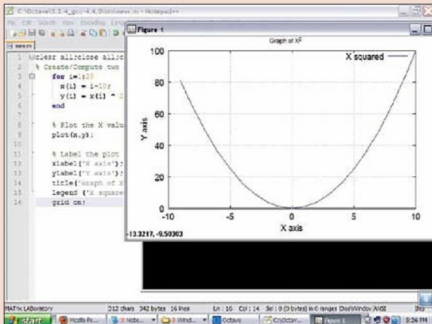


Fig. 5: Creating plots in GNU Octave

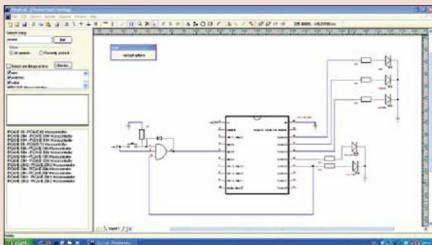


Fig. 6: Designing circuit in TinyCAD

The software consists of a drag-and-drop interface, an interactive wiring system, a circuit time-step control, user-friendly circuit navigation and an export feature that allows users to export the designs to bitmap functions. The program is multi-threaded with a logic simulation engine and a user interface. The UI is built with wxWidgets and the logic gate library, and user-designed circuits are saved in a tag markup language.

## OpenSc

OpenSc is a multi-platform Open Source software that allows design and characterisation of electronic linear cir-

cuits, using either a command-line interface or a GUI. It is similar to proprietary circuit design software and provides an easy integration in self-learning and self-evaluating environments.

## Fendi

Fendi is short for 'Free Electricity Network Drawing Interface.' It is a free graphical user interface used to draw electrical systems. It can be used in simulation engines for operations such as load-flow or short-circuit calculations, stability and transient simulations. ●

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